

The 5 Projects That Actually Get You Hired As A Quant in 2026

Every candidate knows Black-Scholes and Bayes theorem.
Buyside desks hire candidates who build production engines.

01 // MARKET MAKING

L2 Order Book & Matching

Lock-free C++20 engine & queue sim

02 // VOLATILITY

Arbitrage-Free SVI Surface

SSVI & Dupire local vol inversion

03 // STAT ARB

Kalman Filter Pairs Trading

Cointegration & dynamic hedge ratio

04 // RISK & XVA

Multi-Curve OIS & XVA

SOFR dual-curve & American MC

SWIPE FOR ARCHITECTURES >>

L2 Limit Order Book & Matching Engine

Language: Modern C++20 | Focus: Microstructure & Low Latency

AMATEUR GITHUB PROJECT

Downloaded daily AAPL data with yfinance, tested 20/50 SMA crossover in Python.

Why it fails: Immediate auto-reject. Shows zero understanding of order matching, queue priority, exchange latency, or adverse selection.

INDUSTRY-GRADE STANDARD

What top market makers (Citadel, Optiver, Jump) look for:

- Price-Time Priority Engine: Doubly linked lists with $O(1)$ order cancellation via hash map.
- Lock-Free Ingestion: SPSC ring buffer avoiding mutex contention on fast feeds.
- Queue Position Tracking: Model volume ahead of you; don't assume naive touch-fills.
- Microsecond Latency Injector: Pessimistic wire-to-wire ack offsets to prove real PnL.

DESK INTERVIEW DEFENSE QUESTION:

Your limit order sits at the inside bid. 40 lots trade at bid. Did you get filled?

Explain how your matching engine models adverse selection during book sweeps.

Arbitrage-Free SVI Volatility Surface

Language: Python + C++ | Focus: Dupire Local Vol & Static Arbitrage

AMATEUR GITHUB PROJECT

Scraped Yahoo Finance option chains and fitted cubic splines across implied vol.

Why it fails: Standard splines violate no-arbitrage bounds, producing negative butterfly probability densities and imaginary Dupire local volatilities.

INDUSTRY-GRADE STANDARD

What volatility desks look for in your code:

- Gatheral's Quasi-Explicit SVI / SSVI parameterization across strike & expiry.
- Static Arbitrage Free: Strict check that $\partial w / \partial T \geq 0$ (no calendar arbitrage).
- Breeden-Litzenberger Check: $\partial^2 C / \partial K^2 \geq 0$ to guarantee positive risk-neutral density.
- Dupire Local Vol Inversion: Smooth PDE pricing grid for exotic autocallables & barriers.

DESK INTERVIEW DEFENSE QUESTION:

Under what conditions does SSVI surface calibration produce butterfly arbitrage?

How does your optimizer enforce Lee's moment formula at extreme wings?

Stat Arb & Kalman Filter Pairs Trading

Language: Python | Focus: Cointegration, OU Fitting & Slippage

AMATEUR GITHUB PROJECT

Calculated Pearson correlation between KO and PEP, traded when Z-score > 2.

Why it fails: Correlation $\neq$ Cointegration. High correlation between non-stationary prices is often spurious. Unhedged spreads drift apart and blow up the fund.

INDUSTRY-GRADE STANDARD

What quantitative equity / statistical arbitrage teams look for:

- Cointegration Testing: Augmented Dickey-Fuller (ADF) & Johansen eigenvector rank.
- Dynamic Beta Tracking: State-space Kalman filter updating β_t dynamically.
- Ornstein-Uhlenbeck (OU) Calibration: Estimate true half-life of mean reversion.
- Frictions & Half-Spread: Square-root market impact model to weed out phantom alpha.

DESK INTERVIEW DEFENSE QUESTION:

Why does rolling OLS fail for hedge ratio estimation compared to Kalman filter?

How do you tune process noise covariance Q to avoid lagging regime changes?

Multi-Curve OIS Bootstrapping & XVA

Focus: Modern SOFR Discounting & American Monte Carlo (AMC)

PROJECT 04: MULTI-CURVE SOFR

The Post-LIBOR Dual-Curve Architecture:

- Disconnect between discounting curve (SOFR/OIS) and projection curve (Term SOFR).
- Monotone Convex / Cubic Hermite Spline interpolation preventing negative forward rates.
- Jacobian solving: Simultaneous multi-instrument root finding for IRS and basis swaps.

Desk takeaway: Single-curve discounting is obsolete. Desks test dual-curve mastery.

PROJECT 05: XVA SIMULATION ENGINE

Counterparty Credit Risk Under Longstaff-Schwartz:

- American Monte Carlo (AMC): Regression-based exercise boundary across 10k paths.
- Profile generation: Expected Exposure (EE) and 99% Potential Future Exposure (PFE).
- CSA Modeling: Netting sets, collateral threshold, and Margin Period of Risk (MPOR).

Desk takeaway: Risk quants must model netting and default intensity jump curves.

How do you calculate CVA on a netting set with daily margin calls vs. uncollateralized?

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